Programming for Professional Research Using R

Session 3

February 19, 2026

Today

- Learn how to:
  - Filter, mutate, group, and summarize data using Tidyverse functions
  - Reshape data using Tidyverse functions
- Be introduced to:
  - The concept of a "tidy" dataset
- Practice the above!

Data Wrangling

Tidyverse Introduction

Base R Layout | Tidyverse Layout

```r
names(iris)
```

```
## [1] "Sepal.Length" "Sepal.Width"  "Petal.Length" "Petal.Width"  "Species"
```

```r
str_replace(str_to_lower(names(iris)), "\\.", "_")
```

```
## [1] "sepal_length" "sepal_width"  "petal_length" "petal_width"  "species"
```

Tidyverse Introduction

Base R Layout | Tidyverse Layout

Tidyverse functions introduce a 'cleaner' method to write code out, using what is called the 'pipe operator': **%>%**. It's almost like writing a recipe, step by step.

```
names(iris)
```

```
## [1] "Sepal.Length" "Sepal.Width"  "Petal.Length" "Petal.Width"  "Species"
```

```
iris %>%
  names() %>%
  str_to_lower() %>%
  str_replace("\\.", "_")
```

```
## [1] "sepal_length" "sepal_width"  "petal_length" "petal_width"  "species"
```

(Note — you can also write the pipe operator as **|>**)

Basic Wrangling Functions

filter() select() mutate()

`filter()` is used to **extract** rows (a.k.a. observations) from a dataset. It does so using a logical condition.

```
filter_example <- mtcars %>%
  filter(wt > 3)

filter_example %>% head()
```

```
##                    mpg cyl  disp  hp drat    wt  qsec vs am gear carb
## Hornet 4 Drive    21.4   6 258.0 110 3.08 3.215 19.44  1  0    3    1
## Hornet Sportabout 18.7   8 360.0 175 3.15 3.440 17.02  0  0    3    2
## Valiant           18.1   6 225.0 105 2.76 3.460 20.22  1  0    3    1
## Duster 360        14.3   8 360.0 245 3.21 3.570 15.84  0  0    3    4
## Merc 240D         24.4   4 146.7  62 3.69 3.190 20.00  1  0    4    2
## Merc 230          22.8   4 140.8  95 3.92 3.150 22.90  1  0    4    2
```

Basic Wrangling Functions

filter() **select()** mutate()

`select()` is used to **extract** columns (a.k.a variables) from a dataset. It does so using the variable(s)'s name.

```
select_example <- mtcars %>%
  select(
    mpg, carb
  )
select_example %>% head()
```

```
##                    mpg carb
## Mazda RX4         21.0    4
## Mazda RX4 Wag     21.0    4
## Datsun 710        22.8    1
## Hornet 4 Drive    21.4    1
## Hornet Sportabout 18.7    2
## Valiant           18.1    1
```

Basic Wrangling Functions

filter() select() **mutate()**

`mutate()` can be used to either **create** a new column (a.k.a. variable) or to **modify** an existing column (a.k.a. variable).

```
mutate_example <- mtcars %>%
  mutate(
    heavy = case_when(
      wt > 3 ~ "Yes",
      TRUE   ~ "No"
    )
  )
mutate_example %>% select(wt, heavy) %>% head()
```

```
##                     wt heavy
## Mazda RX4          2.620    No
## Mazda RX4 Wag      2.875    No
## Datsun 710         2.320    No
## Hornet 4 Drive     3.215   Yes
## Hornet Sportabout 3.440   Yes
## Valiant            3.460   Yes
```

Basic Wrangling Functions

group_by() and summarize() | pivot_longer() | pivot_longer() result | pivot_wider() | pivot_wider() result

`group_by()` and `summarize()` are used to **aggregate** data, i.e. to summarize information to a different level of observation.

```
group_by_summarize_example <- mtcars %>%
  group_by(cyl) %>%
  summarize(
    mpg = mean(mpg, na.rm = TRUE)
  )

group_by_summarize_example
```

```
## # A tibble: 3 × 2
##     cyl   mpg
##   <dbl> <dbl>
## 1     4  26.7
## 2     6  19.7
## 3     8  15.1
```

Basic Wrangling Functions

group_by() and summarize() | **pivot_longer()** | pivot_longer() result

pivot_wider() | pivot_wider() result

```r
relig_income[1:6] %>% head(n = 2)
```

```
## # A tibble: 2 × 6
##   religion `<$10k` `$10-20k` `$20-30k` `$30-40k` `$40-50k`
##   <chr>      <dbl>     <dbl>     <dbl>     <dbl>     <dbl>
## 1 Agnostic      27        34        60        81        76
## 2 Atheist       12        27        37        52        35
```

```r
relig_income_long <- relig_income %>%
    pivot_longer(
        cols      = !religion, # Everything but religion
        names_to  = "levels",
        values_to = "num"
    )
```

Basic Wrangling Functions

group_by() and summarize() | pivot_longer() | **pivot_longer() result** | pivot_wider() | pivot_wider() result

```
relig_income[1:6] %>% head(n = 2)
```

```
## # A tibble: 2 × 6
##   religion `<$10k` `$10-20k` `$20-30k` `$30-40k` `$40-50k`
##   <chr>      <dbl>     <dbl>     <dbl>     <dbl>     <dbl>
## 1 Agnostic      27        34        60        81        76
## 2 Atheist       12        27        37        52        35
```

```
relig_income_long %>% head(n = 4)
```

```
## # A tibble: 4 × 3
##   religion levels    num
##   <chr>    <chr>   <dbl>
## 1 Agnostic <$10k      27
## 2 Agnostic $10-20k    34
## 3 Agnostic $20-30k    60
## 4 Agnostic $30-40k    81
```

Basic Wrangling Functions

group_by() and summarize() | pivot_longer() | pivot_longer() result

pivot_wider() | pivot_wider() result

```
fish_encounters %>% head(n = 4)
```

```
## # A tibble: 4 × 3
##   fish  station  seen
##   <fct> <fct>   <int>
## 1 4842  Release     1
## 2 4842  I80_1       1
## 3 4842  Lisbon      1
## 4 4842  Rstr        1
```

```
fish_encounters_wide <- fish_encounters %>%
    pivot_wider(
        names_from  = station,
        values_from = seen
    )
```

Basic Wrangling Functions

group_by() and summarize() | pivot_longer() | pivot_longer() result

pivot_wider() | pivot_wider() result

```
fish_encounters %>% head(n = 4)
```

```
## # A tibble: 4 × 3
##   fish  station  seen
##   <fct> <fct>   <int>
## 1 4842  Release     1
## 2 4842  I80_1       1
## 3 4842  Lisbon      1
## 4 4842  Rstr        1
```

```
fish_encounters_wide[1:6] %>% head(n = 2)
```

```
## # A tibble: 2 × 6
##   fish  Release I80_1 Lisbon  Rstr Base_TD
##   <fct>   <int> <int>  <int> <int>   <int>
## 1 4842        1     1      1     1       1
## 2 4843        1     1      1     1       1
```

Working with 'Tidy' Datasets

```
table1
#> # A tibble: 6 x 4
#>   country      year  cases population
#>   <chr>       <int>  <int>      <int>
#> 1 Afghanistan  1999    745   19987071
#> 2 Afghanistan  2000   2666   20595360
#> 3 Brazil       1999  37737  172006362
#> 4 Brazil       2000  80488  174504898
#> 5 China        1999 212258 1272915272
#> 6 China        2000 213766 1280428583
```

```
table2
#> # A tibble: 12 x 4
#>   country      year type           count
#>   <chr>       <int> <chr>          <int>
#> 1 Afghanistan  1999 cases            745
#> 2 Afghanistan  1999 population  19987071
#> 3 Afghanistan  2000 cases           2666
#> 4 Afghanistan  2000 population  20595360
#> 5 Brazil       1999 cases          37737
#> 6 Brazil       1999 population 172006362
#> # … with 6 more rows
```

```
table3
#> # A tibble: 6 x 3
#>   country      year rate
#> * <chr>       <int> <chr>
#> 1 Afghanistan  1999 745/19987071
#> 2 Afghanistan  2000 2666/20595360
#> 3 Brazil       1999 37737/172006362
#> 4 Brazil       2000 80488/174504898
#> 5 China        1999 212258/1272915272
#> 6 China        2000 213766/1280428583

# Spread across two tibbles
table4a  # cases
#> # A tibble: 3 x 3
#>   country     `1999` `2000`
#> * <chr>        <int>  <int>
#> 1 Afghanistan    745   2666
#> 2 Brazil       37737  80488
#> 3 China       212258 213766
table4b  # population
#> # A tibble: 3 x 3
#>   country         `1999`     `2000`
#> * <chr>            <int>      <int>
#> 1 Afghanistan   19987071   20595360
#> 2 Brazil       172006362  174504898
#> 3 China       1272915272 1280428583
```

These are all useable versions of the same data. Only one of them, however, is 'tidy'.

What makes a dataset 'tidy'? From Hadley Wickham, Mine Çetinkaya-Rundel, and Garrett Grolemund, *R for Data Science (2e)* Chapter 5 — Tidy Tidying:

1. Each **variable** must have **its own column**.
2. Each **observation** must have **its own row**.
3. Each **value** must have **its own cell**.

Easier to think about when these conditions are *not* met:

- When one variable is spread across multiple columns.
- When one observation is scattered across multiple rows.

`table1` is the tidy version of this dataset. How can we convert the other versions to be tidy?

table2 table3 table4 table4 results

```r
table2 %>% head(n = 2)
```

```
## # A tibble: 2 × 4
##   country      year type          count
##   <chr>       <dbl> <chr>         <dbl>
## 1 Afghanistan  1999 cases           745
## 2 Afghanistan  1999 population 19987071
```

```r
table2 %>%
  pivot_wider(
    names_from  = type,
    values_from = count
  ) %>% head(n = 4)
```

```
## # A tibble: 4 × 4
##   country      year cases population
##   <chr>       <dbl> <dbl>      <dbl>
## 1 Afghanistan  1999   745   19987071
## 2 Afghanistan  2000  2666   20595360
## 3 Brazil       1999 37737  172006362
## 4 Brazil       2000 80488  174504898
```

table2 | **table3** | table4 | table4 results

```r
table3 %>% head(n = 2)
```

```
## # A tibble: 2 × 3
##   country      year rate
##   <chr>       <dbl> <chr>
## 1 Afghanistan  1999 745/19987071
## 2 Afghanistan  2000 2666/20595360
```

```r
table3 %>%
  mutate(
    cases      = as.numeric(str_extract(rate, ".*(?=\\/)")),
    population = as.numeric(str_extract(rate, "(?⇐\\/).*"))
  ) %>% select(-rate) %>% head(n = 4)
```

```
## # A tibble: 4 × 4
##   country      year cases population
##   <chr>       <dbl> <dbl>      <dbl>
## 1 Afghanistan  1999   745   19987071
## 2 Afghanistan  2000  2666   20595360
## 3 Brazil       1999 37737  172006362
## 4 Brazil       2000 80488  174504898
```

table2 table3 **table4** table4 results

```r
table4a %>% head(n = 2)
```

```
## # A tibble: 2 × 3
##   country     `1999` `2000`
##   <chr>        <dbl>  <dbl>
## 1 Afghanistan    745   2666
## 2 Brazil       37737  80488
```

```r
table4b %>% head(n = 2)
```

```
## # A tibble: 2 × 3
##   country        `1999`    `2000`
##   <chr>           <dbl>     <dbl>
## 1 Afghanistan  19987071  20595360
## 2 Brazil      172006362 174504898
```

table2 table3 table4 **table4 results**

```r
table4a %>%
  pivot_longer(
    cols      = c(`1999`, `2000`),
    names_to  = "year",
    values_to = "cases"
  ) %>%
  left_join(
    table4b %>%
      pivot_longer(
        cols      = c(`1999`, `2000`),
        names_to  = "year",
        values_to = "population"
      )
  ) %>% head(n = 4)
```

```
## Joining with `by = join_by(country, year)`
```

```
## # A tibble: 4 × 4
##   country     year  cases population
##   <chr>       <chr> <dbl>      <dbl>
## 1 Afghanistan 1999    745   19987071
## 2 Afghanistan 2000   2666   20595360
## 3 Brazil      1999  37737  172006362
## 4 Brazil      2000  80488  174504898
```

Practical Exercise — Using the World Values Survey Dataset

The World Values Survey

> *"The survey, which started in 1981, seeks to use the most rigorous, high-quality research designs in each country. The WVS consists of nationally representative surveys conducted in almost 100 countries which contain almost 90 percent of the world's population, using a common questionnaire. [...] WVS seeks to help scientists and policy makers understand changes in the beliefs, values and motivations of people throughout the world."*

- Social values, attitudes & stereotypes
- Societal well-being
- Social capital, trust and organizational membership
- Economic values
- Corruption
- Migration
- Post-materialist index
- Science & technology
- Religious values
- Security
- Ethical values & norms
- Political interest and political participation
- Political culture and political regimes
- Demography

Today's practical component

1. Successfully run the code in the `session_3.R` script
2. Create your own script and do the following:
    - Find mean values for 'importance in life' variables (Q1-6) for countries in another region than Europe
    - Calculate average 'enthusiasm' for these life subjects in countries in another region than Europe
    - Perform the same analysis, either on European countries or other countries, for another group of indicators in the dataset:
        - Important child qualities: Q7-17
        - Neighbors: Q18-26
        - Statements to agree with: Q27-41

NOTE — You should refer to documentation for the dataset, which can be found on Canvas.

Links

Dominic Royé, **“A very short introduction to Tidyverse”**

tidyr, **“Pivoting”**

Hadley Wickham, Mine Çetinkaya-Rundel & Garrett Grolemund, **R for Data Science, 2e**

RStudio, **RStudio Cheatsheets**